NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS3 SUBJECT: Biology

TOPIC: Genetics: Mendelian Inheritance

Introduction

- Genetics is the study of heredity — how characteristics pass from parents to offspring. Gregor Mendel, the father of genetics, worked with garden peas and discovered the laws of inheritance.

Key terms

- **Gene** — a section of DNA controlling one characteristic; **allele** — an alternative form of a gene (T for tall, t for short).
- **Genotype** — the genetic make-up (TT, Tt, tt); **phenotype** — the observable appearance (tall, short).
- **Dominant** — an allele that expresses itself even when paired with a recessive one; **recessive** — expressed only when two copies are present (homozygous).
- **Homozygous** — two identical alleles (TT or tt); **heterozygous** — two different alleles (Tt).

Mendel's laws

- **Law of segregation** — the two alleles of each gene separate during gamete formation, so each gamete carries one allele.
- **Law of independent assortment** — alleles of different genes are inherited independently and combine at random (for genes on different chromosomes).

Monohybrid cross

- Tall (TT) x dwarf (tt) → all F1 tall (Tt). Self the F1 → F2 ratio 3 tall : 1 dwarf (genotype 1 TT : 2 Tt : 1 tt). A test cross (with a homozygous recessive) reveals whether a dominant phenotype is TT or Tt.

CLASS EVALUATION

1. Define gene, allele, genotype and phenotype.
2. State the law of segregation.
3. In a monohybrid cross Tt x Tt, give the genotypic and phenotypic ratios.
4. What is a test cross and when is it used?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: TWO

CLASS: SS3 SUBJECT: Biology

TOPIC: Genetic Crosses & Heredity Problems

Introduction

- Solving heredity problems requires a Punnett square — a grid that shows all possible combinations of parental gametes. This week practises monohybrid and dihybrid crosses, including blood groups.

Steps in solving

- 1. Write the parental genotypes. 2. Work out the gametes of each parent. 3. Draw the Punnett square. 4. Fill the boxes. 5. Read off the genotypic and phenotypic ratios.

Worked example

- In guinea pigs, black fur (B) dominates white (b). A heterozygous black male (Bb) mates with a white female (bb). Gametes: B, b x b, b. Offspring: Bb, Bb, bb, bb → 1 black : 1 white (1:1). If both parents are Bb → 3 black : 1 white.

Blood groups (ABO)

- Alleles IA and IB are codominant; i is recessive. Genotypes: IAIA or IAi (group A); IBIB or IBi (group B); IAIB (group AB); ii (group O). A group O parent (ii) and group AB parent (IAIB) can produce A and B children only.

Sex determination

- Human females are XX, males XY. The male produces X and Y gametes in equal numbers; a cross shows a 1:1 ratio of girls to boys — the father's gamete decides the sex of the child.

CLASS EVALUATION

1. A tall pea plant (Tt) is crossed with a dwarf plant (tt). Show the Punnett square and state the phenotypic ratio.
2. Can two group A parents have a group O child? Explain with a cross.
3. Who determines the sex of a child and how?
4. Distinguish between monohybrid and dihybrid inheritance.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: THREE

CLASS: SS3 SUBJECT: Biology

TOPIC: DNA, RNA & Protein Synthesis

Introduction

- Deoxyribonucleic acid (DNA) is the genetic material of living cells. It carries the instructions for making proteins — the molecules that do most of the work in the cell.

Structure of DNA

- DNA is a double helix of two strands wound round each other. Each strand is a chain of nucleotides; a nucleotide has a phosphate group, a sugar (deoxyribose) and a nitrogenous base. Bases pair specifically (complementary base pairing): **A–T** and **C–G** (in DNA; in RNA A pairs with U). A gene is a sequence of bases coding for a protein.

DNA replication

- The double helix unwinds; each strand acts as a template; new complementary bases are added by enzymes so two identical DNA molecules are formed. This happens before cell division and explains how genetic information is copied exactly.

RNA and protein synthesis

- RNA is single-stranded with the sugar ribose and uracil instead of thymine. **Transcription**: in the nucleus, a messenger RNA (mRNA) copy of one gene is made. **Translation**: the mRNA moves to a ribosome in the cytoplasm, where transfer RNA (tRNA) brings amino acids in the order coded — groups of three bases (codons) specify each amino acid. The chain folds into a protein.

CLASS EVALUATION

1. Describe the structure of DNA.
2. What is complementary base pairing?
3. Distinguish between transcription and translation.
4. How does a gene determine a protein?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS3 SUBJECT: Biology

TOPIC: Genetic Engineering & Biotechnology

Introduction

- Genetic engineering is the artificial manipulation of an organism's genes to produce desired traits — inserting, deleting or modifying DNA. Biotechnology uses living organisms or their products for human benefit.

Tools and techniques

- **Restriction enzymes** cut DNA at specific points; **ligase** joins DNA fragments. **Plasmids** (small circular bacterial DNA) are used as vectors (carriers). **PCR** (polymerase chain reaction) makes millions of copies of a DNA fragment; **electrophoresis** separates DNA fragments by size, used in DNA fingerprinting.

Applications

- **Medicine** — human insulin for diabetics (made by bacteria), vaccines, gene therapy, screening for inherited diseases.
- **Agriculture** — genetically modified (GM) crops resistant to pests, drought or herbicides (Bt maize), and improved yields.
- **Industry** — enzymes in detergents, fermentation for bread, beer and antibiotics (penicillin from Penicillium).
- **Forensics** — DNA fingerprinting identifies criminals and proves paternity.

Ethical concerns

- Safety of GM foods, effects on biodiversity, gene patents, 'designer babies', privacy of genetic information, and the risk of biological weapons. Society needs laws and ethics to guide the technology.

CLASS EVALUATION

1. Define genetic engineering.
2. Name the enzymes used in gene splicing.
3. How is human insulin produced today?
4. State two applications of DNA fingerprinting.

5. Mention two ethical concerns of genetic engineering.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS3 SUBJECT: Biology

TOPIC: Variation & Evolution

Introduction

- Variation is the differences that exist between individuals of the same species. Evolution is the gradual change in living things over generations, from simple to more complex forms.

Types of variation

- **Continuous** — a smooth range of small differences (height, weight, skin colour, shoe size), controlled by many genes and influenced by the environment.
- **Discontinuous** — distinct categories with no intermediates (blood groups, tongue rolling, presence of ear lobes), controlled by one or few genes.

Causes of variation

- Genetic: mutation, crossing over during meiosis, random fertilisation, independent assortment. Environmental: nutrition, climate, exercise, light. Most variation results from both heredity and environment.

Evolution by natural selection (Darwin)

- Individuals in a population show variation; more offspring are produced than can survive; there is competition for resources (struggle for existence); individuals with favourable variations survive, reproduce and pass the trait on (survival of the fittest); over many generations the species changes. Evidence for evolution: fossils, comparative anatomy, vestigial organs, embryology and molecular similarities. Lamarck proposed the inheritance of acquired characteristics — now rejected.

CLASS EVALUATION

1. Differentiate between continuous and discontinuous variation with examples.
2. Name two causes of variation.
3. Outline Darwin's theory of natural selection.
4. Give three evidences of evolution.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: SIX

CLASS: SS3 SUBJECT: Biology

TOPIC: Adaptation

Introduction

- Adaptation is the possession of structures or behaviours that enable an organism to survive and reproduce in its environment. Adaptation arises from natural selection acting on variation.

Types of adaptation

- **Structural** — body features: the camel's hump (fat store), long legs of the heron (wading), webbed feet of ducks (swimming), flat teeth of herbivores (grinding), spines of cacti (water loss), gills of fish (breathing in water).
- **Behavioural** — actions: migration of birds, hibernation, aestivation, nocturnal activity, herding, protective display, parental care.
- **Physiological** — internal processes: production of antifreeze by arctic fish, salt glands of sea birds, camouflage (cryptic coloration) and mimicry as protective adaptations.

Adaptation to specific environments

- **Xerophytes** (desert): thick cuticle, sunken stomata, reduced leaves, water storage, long roots. **Hydrophytes** (water): large air spaces, floating leaves, reduced roots. **Halophytes** (salt): salt toleration. **Desert animals**: nocturnal, efficient kidneys, humps in camels, burrowing.

Adaptation and evolution

- Adaptations are the visible results of natural selection — the environment 'selects' the traits that work. They help explain biodiversity and why species occupy particular niches.

CLASS EVALUATION

1. Define adaptation.
2. Give two structural adaptations of a fish and a cactus.
3. Distinguish between structural and behavioural adaptation.
4. Why do deserts have few but highly adapted species?